Jared Duong

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Education

University of Guelph **Sep 2024 – Apr 2029**
Bachelor of Engineering in Computer Engineering (Co-op) *Guelph, ON*

Technical Skills

Programming Languages: C, C++, Python, VHDL, MATLAB, Java, C#, JavaScript, SQL, HTML/CSS
Hardware & Embedded: Arduino, FPGA, Digital Logic, PCB Design, Sensor Integration, Motor Control, Soldering
Engineering Tools: SOLIDWORKS (CSWA), AutoCAD, Fusion 360, ModelSim, Vivado, Git, GitHub, 3D Printing

Experience

Software Integration Developer **Jan 2026 – Mar 2026**
Appleby North Paint Inc. *Remote*

- Developed a Python data-processing pipeline using pandas and openpyxl to process product and inventory exports, generating structured feeds for **206 SKUs**.
- Implemented automated matching across three datasets, linking **154 products** to pricing records and identifying **52 unmatched SKUs**.
- Automated inventory classification and generated diagnostic reports identifying **146 pricing discrepancies** for investigation.

Summer Production Student **May 2025 – Jun 2025**
Hartmann Canada Inc. *Brantford, ON*

- Operated and monitored manufacturing equipment, inspecting control-panel indicators and gauges to identify potential equipment and production issues.
- Performed quality-control inspections, verifying product weight, colour, and functionality against established manufacturing specifications.
- Documented inspection observations, reported production concerns, and followed workplace safety and operating procedures.

Projects

MIPS-Based 16-Bit CPU System | *VHDL, ModelSim, Vivado* **Jan 2026 – Mar 2026**

- Co-developed a **16-bit single-cycle CPU** in VHDL integrating an ALU, **16-register** file, hardwired control unit, and instruction/data memories.
- Implemented **10 instructions** supporting arithmetic, logic, memory access, branching, and jumping operations.
- Verified digital logic and instruction execution using ModelSim testbenches and waveform analysis, troubleshooting control and write-back logic.

4x4x4 LED Cube with Custom PCB | *Arduino, C/C++, PCB Design* **Mar 2025 – Apr 2025**

- Designed and soldered a **64-LED cube** using 74HC04 and SN74HC08 logic ICs, a custom PCB, and a 3D-printed base.
- Developed Arduino Nano control logic in C/C++ using multiplexing, layer scanning, and timing to drive programmable LED patterns.

Autonomous Wheelchair Prototype | *Arduino, C++, SOLIDWORKS* **Sep 2024 – Dec 2024**

- Developed an Arduino-based wheelchair prototype integrating sensors and motor controls for autonomous navigation.
- Programmed and tested sensor-driven navigation and motor-control behaviour in C++ during prototype operation.
- Performed centre-of-mass and tipping analysis and refined the mechanical design to improve stability and reduce component costs.

Engineering Involvement

Mechanical Team Member **Sep 2024 – Present**
University of Guelph Robotics Team *Guelph, ON*

- Designing a camera mount in SOLIDWORKS for a competition rover, focusing on component fit, mounting geometry, and mechanical integration.
- Assisted with drivetrain and wheel-subsystem assembly and collaborated with electrical and software subteams on integration, design reviews, and testing.